Roll No. ....

TEC–301

B. TECH. (ECE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

ELECTRONIC DEVICES AND CIRCUITS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain BJT as switch and its switching timings. (CO1)

OR

(b) Explain the terms I_{CEO} and I_{CBO} with respect to leakage current. A BJT has $\alpha = 0.97$ and $I_{CO} = 50$ nA and is operating at base current of $10\,\mu$A, find I_C, I_E, I_{CEO} and I_{CBO}. (CO1)

2. (a) Explain the following terms : (CO1 & CO2)

(i) Emitter Injection Efficiency

(ii) Base Transport Factor

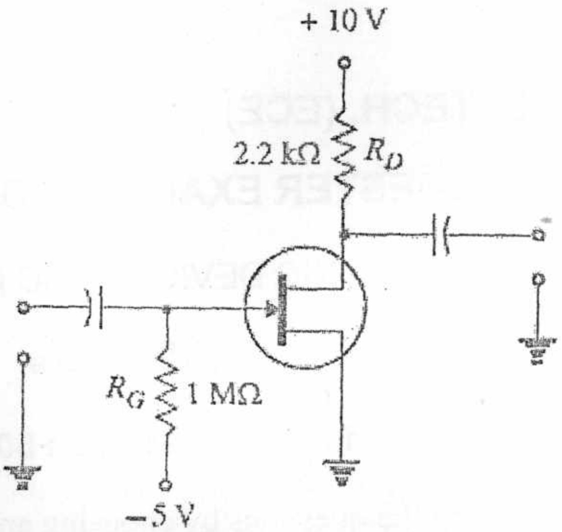
(iii) Punch Through

P. T. O.

OR

(b) A JFET shown in figure has values of $V_{GS\,(off)} = -8$ V and $I_{DSS} = 16$ mA. Determine the values of V_{GS}, I_D and V_{DS} for the circuit.

(CO1 & CO2)



3. (a) Describe signal variation and explain amplifier classification based on signal variation. (CO1 & CO2)

OR

(b) Explain the working of *n*-channel JFETs in detail with proper diagrams. A JFET has the following parameters :

$$I_{DSS} = 32 \text{ mA};\ V_{GS\,(off)} = -8\text{V},\ V_{GS} = -4.5 \text{ V}.$$

Find the value of drain current. (CO1 & CO2)

4. (a) Explain the input and output characteristics of CB configuration with proper diagrams. (CO1 & CO2)

OR

(b) Discuss the transfer and output characteristics of E-only MOSFET.

(CO1 & CO2)

5. (a) Discuss the Darlington pair configuration for BJT amplifier. (CO2)

OR

(b) Figure shows the biasing with base resistor method. (CO2)

(i) Determine the collector current I_C and collector-emitter voltage V_{CE}. Given that $\beta = 50$.

(ii) If R_B in this circuit is changed to 50 $k\Omega$, find the new operating point.